

Test Summary Report

Restful-Booker (Hotel Booking API)

1. Test Objective

The objective of this testing effort was to validate the functional correctness, data validation, authentication behavior, and non-functional characteristics (response time, headers) of the Restful-Booker API, a hotel booking demo API used for practicing and demonstrating API testing skills. Testing was designed against a formal requirements document and executed using Postman.

2. Scope

The following endpoints and behaviors were covered:

- POST /auth - token generation (valid and invalid credentials)
- POST /booking - booking creation, including required field, data type, and business-logic validation
- GET /booking and GET /booking/{id} - retrieval, filtering, and error handling
- PUT /booking/{id} - full update, with and without authentication
- PATCH /booking/{id} - partial update, with and without authentication
- DELETE /booking/{id} - deletion, authentication enforcement, and repeat-delete (idempotency) behavior
- GET /ping - health check
- Non-functional checks - response time and Content-Type header validation

3. Test Environment

Item	Detail
Tool	Postman
Base URL	https://restful-booker.herokuapp.com
Test Type	Manual functional API testing (positive, negative, boundary, security, non-functional)
Test Basis	Restful_Booker_API_Requirements.docx
Tester	Adedeji Adetoun
Test Period	July 2026

4. Test Execution Summary

Metric	Count	Percentage
Total Test Cases	30	100%
Passed	24	80%
Failed (root cause: real defects)	6	20%
Failed then resolved on retest (stale test data, not defects)	6 of the above 30 were initially flagged Fail and later confirmed Pass after retesting with fresh data	-

Note: during execution, 6 test cases (TC_RB_018, 021, 022, 023, 025, 027) initially returned unexpected 403 Forbidden or ambiguous results. Investigation traced this to stale booking IDs and auth tokens carried over from earlier in a long test session, caused by the API's automatic 10-minute data reset - not a product defect. All 6 were retested using freshly created bookings and freshly generated tokens and passed cleanly. See Section 6 for details.

5. Defect Summary

Severity	Open Defects
Critical	0
High	0
Medium	5
Low	3
Total Open	8
Investigated & Closed (not a bug)	2 (BUG_009, BUG_010 - false positives from stale test data, confirmed via retest)

5.1 Confirmed Open Defects

Bug ID	Summary	Endpoint	Severity
BUG_001	POST /booking returns 500 Internal Server Error (not a graceful 400) when a required field is omitted	POST /booking	High
BUG_002	No validation on data type for totalprice - accepts a string value	POST /booking	Medium
BUG_003	No validation enforcing the required bookingdates object	POST /booking	Medium
BUG_004	Negative totalprice values are accepted without validation	POST /booking	Low
BUG_005	Illogical date ranges (checkout before checkin) are accepted without validation	POST /booking	Medium
BUG_006	firstname filter response omits the firstname field, making results unverifiable	GET /booking	Low
BUG_007	Date range filter (checkin/checkout) returns no results despite matching data existing	GET /booking	Medium
BUG_008	Non-numeric booking ID does not return a validation error	GET /booking/{id}	Medium

Full details, steps to reproduce, and evidence for each defect are documented in Restful_Booker_Bug_Log.xlsx.

6. Key Findings

- Restful-Booker performs minimal server-side validation on POST /booking - most invalid inputs (wrong data types, missing nested objects, negative values, illogical dates) are silently accepted rather than rejected, which would be a concern in a production system.
- One genuine reliability issue was found: omitting a required field causes an unhandled 500 Internal Server Error rather than a controlled 400 response (BUG_001).
- Filtering behavior on GET /booking appears unreliable for date ranges, and the firstname filter's response shape makes correctness difficult to verify directly.
- Two apparent authentication failures (403 Forbidden on PUT/PATCH/DELETE) were investigated and traced to stale test data caused by the API's 10-minute reset cycle, not a genuine defect. This was confirmed through a controlled retest using freshly created bookings and tokens, and is documented as a process/testing-environment finding rather than a product bug.
- Documented "quirks" (DELETE returning 201 instead of 200/204, /ping returning 201 instead of 200, and invalid auth credentials returning 200 with an error in the body instead of 401) were all confirmed as observed, consistent behavior rather than intermittent issues.

7. Risks & Recommendations

- Recommend addressing BUG_001 (500 error) as a priority - unhandled server exceptions on invalid input are a reliability and security concern beyond simple validation gaps.
- Recommend adding server-side validation for data types, required nested objects, and logical business rules (e.g. checkout after checkin) on POST /booking.
- Recommend investigating the date-range filter on GET /booking, as it currently does not appear to return matching results.
- For future test cycles against this API, recommend generating fresh test data (bookings and tokens) immediately before each dependent test step, rather than reusing data created earlier in a long session, to avoid false failures caused by the automatic data reset.

8. Conclusion

Overall, the Restful-Booker API's core CRUD and authentication functionality behaves as documented once tests are executed against fresh test data. However, input validation is notably lenient across several endpoints, and one confirmed reliability defect (500 error on missing required field) was identified. As a demo/practice API this is expected and intentional in places, but the same gaps in a production system would represent meaningful risk. Testing is considered complete for the defined scope, with 8 open defects logged for tracking and 2 initial false positives fully investigated, explained, and closed.

9. Related Artifacts

- Restful_Booker_API_Requirements.docx - requirements and expected behavior specification
- Restful_Booker_Test_Cases_30.xlsx - full 30-case test suite with executed results
- Restful_Booker_Bug_Log.xlsx - detailed defect log with steps to reproduce and investigation notes